

Dependent ambiguity sets for the belief-ambiguity bilevel model: complete extension of the results and proofs

Working note, companion to the manuscript
A Distributionally Robust Model for Bilevel Programs under Asymmetric Beliefs

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요약

본 노트는 원고(비제로섬 JOC 판, [paper/joc-non_zero_sum/](#))의 Assumption 4(곱집합 $\mathcal{D} = \hat{\mathcal{D}} \times \tilde{\mathcal{D}}$)를 종속 집합

$$\mathcal{D}_\delta = \{(p, \tilde{p}) \in \hat{\mathcal{D}} \times \tilde{\mathcal{D}} : d_{\text{TV}}(p, \tilde{p}) \leq \delta\}$$

(더 일반적으로 임의의 polytope $\mathcal{D} \subseteq \hat{\mathcal{D}} \times \tilde{\mathcal{D}}$)로 바꿨을 때, 원고의 결과와 증명이 어떻게 확장되는지를 빠짐없이 정리한다. 원고의 기호와 가정 1–3, Lemma 1(pessimistic recourse)은 그대로 쓴다.

원고 결과	종속 판	본 노트
식 (V-response), 분리 해석	응답마다 다른 리더 집합 $\hat{\mathcal{D}}_{\mathcal{D}}(x, \alpha)$	Prop. 1
Lemma (certificate) (i)–(ii)	결합 penalty 로 그대로	Lemma 2
Lemma (certificate) (iii)	새 증명 (유한 꼭짓점 + 단일 belief Hoffman)	Lemma 2
Theorem (minimax)	$\mathfrak{B} \rightarrow \mathfrak{B}_{\mathcal{D}}$, F 불변. 증명은 belief 고정 쌍대로 단순화	Theorem 1
Theorem (finite-type)	type 마다 리더 집합 $\hat{\mathcal{D}}_I$ 가 붙음	Theorem 2
Theorem (Σ_2^p), Corollary	모든 $\varepsilon, \delta > 0$ 에서 성립	Theorem 3
Prop. (belief cuts) (i)–(ii)	그대로	Prop. 3
Prop. (belief cuts) (iii)	증명 수정 (joint 꼭짓점)	Prop. 3
Lemma (price bounds), α -B&B, Prop. (finite termination)	그대로 (Γ 에 결합 행 추가)	§7

새로 증명해야 했던 것은 Lemma 2(iii) 하나이고, 그 증명은 직사각형 경우에도 그대로 적용된다(원고 증명의 Step 1–2를 대체 가능). 나머지는 기계적 수정이다. 반경 민감도 절(supplement, 현재 미사용)은 다루지 않는다.

1 Setting

Throughout, Assumptions 1–3 of the manuscript hold, $\Xi = \{\xi^1, \dots, \xi^S\}$, and $\hat{\mathcal{D}}, \tilde{\mathcal{D}}$ are the total-variation balls of radii $\hat{\varepsilon}, \tilde{\varepsilon}$ centered at the leader’s reference $\hat{q}$ (manuscript eq. (TV-def)). Assumption 4 is replaced by the following.

Assumption 4’ (Dependent ambiguity set). The ambiguity set is a nonempty polytope $\mathcal{D} \subseteq \hat{\mathcal{D}} \times \tilde{\mathcal{D}}$. The leading instance is the *belief-coupled* set

$$\mathcal{D}_\delta := \{(p, \tilde{p}) \in \hat{\mathcal{D}} \times \tilde{\mathcal{D}} : d_{\text{TV}}(p, \tilde{p}) \leq \delta\}, \quad \delta \in [0, 1], \quad (1)$$

which is a polytope through the lift $-\varsigma \leq p - \tilde{p} \leq \varsigma$, $\mathbf{1}^\top \varsigma \leq 2\delta$.

The parameter δ bounds how far the follower's private belief may be from the true distribution; $\delta = 0$ is a common (but ambiguous) prior, and the product set of the manuscript is recovered for $\delta \geq \hat{\varepsilon} + \tilde{\varepsilon}$ (Lemma 1). Another instance is the likelihood-ratio set $\mathcal{D}_\kappa := \{(p, \tilde{p}) \in \hat{\mathcal{D}} \times \tilde{\mathcal{D}} : \kappa \tilde{p} \leq p\}$ (posterior of a private signal of probability at least κ); every result below that is stated for a general polytope $\mathcal{D}$ applies to it verbatim.

For $x \in \mathcal{X}$, write $Q_x(\alpha) := (Q(x, \alpha, \xi^s))_s$, $\phi_x(\alpha) := (\phi(x, \alpha, \xi^s))_s$,

$$\psi_x(\tilde{p}) := \max_{h \in H} \tilde{p}^\top Q_x(h), \quad \text{gap}_x(\tilde{p}, \alpha) := \psi_x(\tilde{p}) - \tilde{p}^\top Q_x(\alpha) \quad (\geq 0),$$

so that $\Psi(x, \tilde{p}) = \{\alpha \in H : \text{gap}_x(\tilde{p}, \alpha) = 0\}$, and the *rationalizing set* $\mathcal{T}_x(\alpha) := \{\tilde{p} \in \Delta^S : \alpha \in \Psi(x, \tilde{p})\}$. The leader's worst-case value is

$$V^*(x) := \sup_{(p, \tilde{p}) \in \mathcal{D}} \sup_{\alpha \in \Psi(x, \tilde{p})} \mathcal{R}^p[\phi(x, \alpha, \xi)]. \quad (2)$$

We use the dual representation of CVaR $_\beta$ on a finite support,

$$\mathcal{R}^p[Z] = \max \{ \nu^\top Z : \nu \in L_\beta(p) \}, \quad L_\beta(p) := \{ \nu : 0 \leq \nu \leq p/(1 - \beta), \mathbf{1}^\top \nu = 1 \}, \quad (3)$$

and the belief polytope

$$\mathfrak{B}_\mathcal{D} := \{(p, \nu, \tilde{p}) : (p, \tilde{p}) \in \mathcal{D}, \nu \in L_\beta(p)\} \quad \left(\text{for } \mathcal{D} = \hat{\mathcal{D}} \times \tilde{\mathcal{D}} \text{ this is the manuscript's } \mathfrak{B} \right). \quad (4)$$

Lemma 1 (Geometry of $\mathcal{D}_\delta$). (a) For $\tilde{p} \in \Delta^S$, $\text{dist}_{\text{TV}}(\tilde{p}, \hat{\mathcal{D}}) = (d_{\text{TV}}(\tilde{p}, \hat{q}) - \hat{\varepsilon})^+$, and symmetrically for $\tilde{\mathcal{D}}$.

(b) $\text{proj}_{\tilde{p}} \mathcal{D}_\delta = \tilde{\mathcal{D}}_{\tilde{\varepsilon}'}$ and $\text{proj}_p \mathcal{D}_\delta = \hat{\mathcal{D}}_{\hat{\varepsilon}'}$ with $\tilde{\varepsilon}' := \min\{\tilde{\varepsilon}, \hat{\varepsilon} + \delta\}$, $\hat{\varepsilon}' := \min\{\hat{\varepsilon}, \tilde{\varepsilon} + \delta\}$, where $\tilde{\mathcal{D}}_r, \hat{\mathcal{D}}_r$ denote the TV balls of radius r around $\hat{q}$.

(c) $\mathcal{D}_\delta = \hat{\mathcal{D}} \times \tilde{\mathcal{D}}$ whenever $\delta \geq \hat{\varepsilon} + \tilde{\varepsilon}$; $\mathcal{D}_\delta$ is nondecreasing in δ ; and $(\hat{q}, \hat{q}) \in \mathcal{D}_\delta$ for every $\delta \geq 0$.

Proof. (a) Let $d := d_{\text{TV}}(\tilde{p}, \hat{q}) > \hat{\varepsilon}$ (otherwise $\tilde{p} \in \hat{\mathcal{D}}$). The point $p := \hat{q} + (\hat{\varepsilon}/d)(\tilde{p} - \hat{q})$ is a convex combination of $\hat{q}$ and $\tilde{p}$, hence in Δ^S , with $d_{\text{TV}}(p, \hat{q}) = \hat{\varepsilon}$ and $d_{\text{TV}}(p, \tilde{p}) = d - \hat{\varepsilon}$. Conversely $d_{\text{TV}}(p', \tilde{p}) \geq d_{\text{TV}}(\tilde{p}, \hat{q}) - d_{\text{TV}}(p', \hat{q}) \geq d - \hat{\varepsilon}$ for every $p' \in \hat{\mathcal{D}}$. (b) $\tilde{p} \in \text{proj}_{\tilde{p}} \mathcal{D}_\delta$ iff $\tilde{p} \in \tilde{\mathcal{D}}$ and $\text{dist}_{\text{TV}}(\tilde{p}, \hat{\mathcal{D}}) \leq \delta$, i.e. by (a) $d_{\text{TV}}(\tilde{p}, \hat{q}) \leq \min\{\tilde{\varepsilon}, \hat{\varepsilon} + \delta\}$. The other projection is symmetric. (c) For $(p, \tilde{p}) \in \hat{\mathcal{D}} \times \tilde{\mathcal{D}}$, $d_{\text{TV}}(p, \tilde{p}) \leq d_{\text{TV}}(p, \hat{q}) + d_{\text{TV}}(\hat{q}, \tilde{p}) \leq \hat{\varepsilon} + \tilde{\varepsilon}$. Monotonicity and $(\hat{q}, \hat{q}) \in \mathcal{D}_\delta$ are immediate. $\square$

2 Response representation

Proposition 1 (Response representation under dependence; replaces (V-response)). *Under Assumptions 1–3 and 4', for every $x \in \mathcal{X}$,*

$$\begin{aligned} V^*(x) &= \max \left\{ \mathcal{R}^p[\phi(x, \alpha, \xi)] : (p, \tilde{p}) \in \mathcal{D}, \alpha \in \Psi(x, \tilde{p}) \right\} \\ &= \max_{\alpha \in \Psi(x, \text{proj}_{\tilde{p}} \mathcal{D})} \max_{p \in \hat{\mathcal{D}}_\mathcal{D}(x, \alpha)} \mathcal{R}^p[\phi(x, \alpha, \xi)], \end{aligned} \quad (5)$$

with all maxima attained, where $\Psi(x, \text{proj}_{\tilde{p}} \mathcal{D}) = \bigcup_{\tilde{p} \in \text{proj}_{\tilde{p}} \mathcal{D}} \Psi(x, \tilde{p})$ and

$$\hat{\mathcal{D}}_\mathcal{D}(x, \alpha) := \{p : \exists \tilde{p} \in \mathcal{T}_x(\alpha) \text{ with } (p, \tilde{p}) \in \mathcal{D}\}$$

is a nonempty convex compact set for every admissible α . For $\mathcal{D} = \mathcal{D}_\delta$: $\Psi(x, \text{proj}_{\tilde{p}} \mathcal{D}_\delta) = \Psi(x, \tilde{\mathcal{D}}_{\tilde{\varepsilon}'})$ and $\hat{\mathcal{D}}_{\mathcal{D}_\delta}(x, \alpha) = \hat{\mathcal{D}} \cap ((\mathcal{T}_x(\alpha) \cap \tilde{\mathcal{D}}) \oplus B_\delta^{\text{TV}})$. For the product set, $\hat{\mathcal{D}}_\mathcal{D}(x, \alpha) = \hat{\mathcal{D}}$ for every admissible α , which is the manuscript's (V-response).

Proof. The first equality is the definition (2) written as one supremum. The feasible set $G_x := \{(\alpha, p, \tilde{p}) : (p, \tilde{p}) \in \mathcal{D}, \alpha \in \Psi(x, \tilde{p})\}$ is compact: $\mathcal{D}$ and H are compact, and the graph of $\tilde{p} \mapsto \Psi(x, \tilde{p})$ is closed because this correspondence is upper hemicontinuous with closed values (Berge's maximum theorem, as in the supplement's section "The response-set representation"). The objective $(\alpha, p) \mapsto \mathcal{R}^p[\phi_x(\alpha)]$ is continuous: ϕ_x is continuous by Lemma 1(iii), and $(p, z) \mapsto \mathcal{R}^p[z]$ is continuous by (3) (a maximum of a bilinear function over the continuous compact-valued correspondence L_β). Hence the maximum is attained. Grouping the variables as α first gives the second expression. The set $\mathcal{T}_x(\alpha) = \{\tilde{p} \in \Delta^S : \psi_x(\tilde{p}) - \tilde{p}^\top Q_x(\alpha) \leq 0\}$ is a closed sublevel set of a convex function (ψ_x is a maximum of linear functions of $\tilde{p}$), hence convex and compact, so $\hat{\mathcal{D}}_{\mathcal{D}}(x, \alpha)$, the projection of the convex compact set $\mathcal{D} \cap (\Delta^S \times \mathcal{T}_x(\alpha))$, is convex and compact. For $\mathcal{D}_\delta$ the two identities follow from Lemma 1(b) and the definition of $\mathcal{D}_\delta$. $\square$

Remark 1 (Interpretation). Under the product set, $\hat{\mathcal{D}}$ enters only the objective and $\tilde{\mathcal{D}}$ only the response set. Under dependence the response set is still determined by the follower side alone (through $\text{proj}_{\tilde{p}} \mathcal{D}$), but the leader-side set becomes response-dependent: observing which responses are rationalizable restricts the plausible truths to $\hat{\mathcal{D}}_{\mathcal{D}}(x, \alpha)$. The sentences of the manuscript that assert the separation (Section 3 opening paragraph after (V-response), the paragraph after Lemma 2, the paragraph after Theorem 2, and the abstract) should be rephrased accordingly.

3 Certificate of the reaction set

For $\lambda^U \in [0, \infty]$ let $\mathfrak{F}_{\lambda^U}(x, \alpha; \tilde{p})$ denote the follower block (VF-def) of the manuscript *at the fixed belief* $\tilde{p}$, i.e. (VF-def) without the supremum over $\tilde{p}$ and with $\lambda \leq \lambda^U$. The proof of the manuscript's Lemma 2 shows, for fixed $\tilde{p}$,

$$\mathfrak{F}_{\lambda^U}(x, \alpha; \tilde{p}) = \inf_{0 \leq \lambda \leq \lambda^U} (-\lambda \text{gap}_x(\tilde{p}, \alpha)) = \begin{cases} -\lambda^U \text{gap}_x(\tilde{p}, \alpha), & \lambda^U < \infty, \\ 0 \text{ if } \text{gap}_x(\tilde{p}, \alpha) = 0, -\infty \text{ otherwise,} & \lambda^U = \infty. \end{cases} \quad (6)$$

Define the coupled penalized value

$$V_{\lambda^U}^*(x) := \sup_{\alpha \in H} \sup_{(p, \tilde{p}) \in \mathcal{D}} \left[\mathcal{R}^p[\phi(x, \alpha, \xi)] + \mathfrak{F}_{\lambda^U}(x, \alpha; \tilde{p}) \right]. \quad (7)$$

For the product set, $\sup_{\tilde{p} \in \tilde{\mathcal{D}}} \mathfrak{F}_{\lambda^U}$ is the manuscript's $\bar{V}_{\lambda^U}^F$, and (7) is the manuscript's $V_{\lambda^U}^*$.

Lemma 2 (Certificate of the reaction set under dependence; replaces Lemma 2). *Under Assumptions 1–3 and 4', for every $x \in \mathcal{X}$:*

- (i) (Exact certificate) $V_\infty^*(x) = V^*(x)$.
- (ii) (Bounded multiplier) For $\lambda^U < \infty$, $V_{\lambda^U}^*(x) = \sup_{\alpha \in H} \sup_{(p, \tilde{p}) \in \mathcal{D}} [\mathcal{R}^p[\phi_x(\alpha)] - \lambda^U \text{gap}_x(\tilde{p}, \alpha)] \geq V^*(x)$, and $V_{\lambda^U}^*(x)$ is nonincreasing in λ^U .
- (iii) (Exact penalty) There is $\bar{\lambda} < \infty$, independent of x , such that $V_{\lambda^U}^*(x) = V^*(x)$ for all $\lambda^U \geq \bar{\lambda}$.

Proof. (i) By (6) with $\lambda^U = \infty$, the bracket in (7) equals $\mathcal{R}^p[\phi_x(\alpha)]$ if $\alpha \in \Psi(x, \tilde{p})$ and $-\infty$ otherwise, so $V_\infty^*(x)$ is (2).

(ii) The formula is (6). The penalty $\lambda^U \text{gap}_x \geq 0$ vanishes on $\{(\alpha, p, \tilde{p}) : (p, \tilde{p}) \in \mathcal{D}, \alpha \in \Psi(x, \tilde{p})\}$, so $V_{\lambda^U}^*(x) \geq V^*(x)$; the penalty is nondecreasing in λ^U , so $V_{\lambda^U}^*(x)$ is nonincreasing.

(iii) Fix x . By (3) and (ii),

$$V_\lambda^*(x) = \sup_{\alpha \in H} \max_{b \in \mathfrak{B}_{\mathcal{D}}} \Phi_\lambda(\alpha, b), \quad \Phi_\lambda(\alpha, b) := \nu^\top \phi_x(\alpha) - \lambda [\psi_x(\tilde{p}) - \tilde{p}^\top Q_x(\alpha)], \quad b = (p, \nu, \tilde{p}). \quad (8)$$

Step 1 (finitely many belief vertices). Let P_x be the lifted polytope of the supplement's proof of Theorem 2, so that $\psi_x(\tilde{p}) = \max\{\tilde{p}^\top \vartheta : (h, \vartheta) \in P_x\}$. P_x is a pointed polyhedron (h ranges over the compact H and ϑ is bounded above), and the maximum is finite for $\tilde{p} \geq 0$, so it is attained at one of the finitely many vertices (h^v, ϑ^v) , $v \in \text{vert}(P_x)$. Hence $\psi_x(\tilde{p}) = \max_v \tilde{p}^\top \vartheta^v$ is convex and piecewise linear, and Δ^S is covered by the finitely many polytopes $C_v := \{\tilde{p} \in \Delta^S : \tilde{p}^\top \vartheta^v \geq \tilde{p}^\top \vartheta^{v'} \forall v'\}$, on each of which $\psi_x(\tilde{p}) = \tilde{p}^\top \vartheta^v$. Put $K_v := \mathfrak{B}_{\mathcal{D}} \cap \{\tilde{p} \in C_v\}$, a polytope, and $V_x := \bigcup_v \text{vert}(K_v)$, a finite set that depends on neither α nor λ . For fixed (α, λ) , $\Phi_\lambda(\alpha, \cdot)$ is linear on each K_v , so its maximum over K_v is attained at a vertex, and since $\mathfrak{B}_{\mathcal{D}} = \bigcup_v K_v$,

$$V_\lambda^*(x) = \max_{b \in V_x} \sup_{\alpha \in H} \Phi_\lambda(\alpha, b).$$

Step 2 (error bound for a single belief). Fix $b = (p, \nu, \tilde{p}) \in V_x$ and put $f_b(\alpha) := \text{gap}_x(\tilde{p}, \alpha)$. By LP duality and Assumption 3, $Q(x, \alpha, \xi^s) = \min_{\sigma \in \mathcal{V}_F} \sigma^\top (b_x(\xi^s) + B\alpha)$ with $\mathcal{V}_F$ the finite vertex set of $\{\sigma \geq 0 : A^\top \sigma \geq d_F\}$. Since $\tilde{p} \geq 0$, $f_b(\alpha) = \psi_x(\tilde{p}) + \sum_s \max_{\sigma \in \mathcal{V}_F} (-\tilde{p}_s \sigma^\top (b_x(\xi^s) + B\alpha)) = \max_{j \in J} (a_j^\top \alpha - c_j)$ for a finite family of affine functions. Moreover $f_b \geq 0$ on H and $Z_b := \{\alpha \in H : f_b(\alpha) \leq 0\} = \Psi(x, \tilde{p}) \neq \emptyset$. Applying Hoffman's error bound to the linear system $\{W\alpha \leq w, \alpha \geq 0, a_j^\top \alpha \leq c_j (j \in J)\}$, whose solution set is Z_b , there is $\kappa_b < \infty$ with

$$\text{dist}(\alpha, Z_b) \leq \kappa_b \max_j (a_j^\top \alpha - c_j)^+ = \kappa_b f_b(\alpha) \quad \forall \alpha \in H,$$

because the rows of H are satisfied at $\alpha \in H$.

Step 3 (Lipschitz objective). Each $\phi(x, \cdot, \xi^s)$ is continuous and piecewise linear on the compact H (Lemma 1(iii)), hence Lipschitz with some constant L_s ; ν is a probability vector, so $\nu^\top \phi_x(\cdot)$ is Lipschitz with constant $L_x := \max_s L_s$.

Step 4 (conclusion). Let $\lambda \geq \bar{\lambda}_x := L_x \max_{b \in V_x} \kappa_b$. For $b \in V_x$ and $\alpha \in H$, let α' be a nearest point of Z_b . Then

$$\Phi_\lambda(\alpha, b) \leq \nu^\top \phi_x(\alpha') + (L_x - \lambda/\kappa_b) \|\alpha - \alpha'\| \leq \nu^\top \phi_x(\alpha') \leq \mathcal{R}^p[\phi_x(\alpha')] \leq V^*(x),$$

where the third inequality is (3) with $\nu \in L_\beta(p)$, and the last holds because $(p, \tilde{p}) \in \mathcal{D}$ and $\alpha' \in \Psi(x, \tilde{p})$. By Step 1, $V_\lambda^*(x) \leq V^*(x)$, and (ii) gives equality. Since $\mathcal{X}$ is finite, $\bar{\lambda} := \max_{x \in \mathcal{X}} \bar{\lambda}_x < \infty$. $\square$

Remark 2. The argument uses the dependence only through the fact that $\mathfrak{B}_{\mathcal{D}}$ is a polytope, so it covers the product set as well and can replace Steps 1–2 of the manuscript's proof (the conjugate representation of G_x and the piecewise error bound). It also explains why the original route does not extend directly: there the follower's belief is eliminated first ($G_x = \min_{\tilde{p}} \text{gap}_x$), which is impossible when the objective depends on p and p is tied to $\tilde{p}$; here the belief is fixed first at one of finitely many vertices, after which only a single-belief error bound is needed.

4 Minimax reformulation

Let $\Omega_{\mathcal{D}}$ be the set Ω of the manuscript (eq. (Omega)) with $\mathfrak{B}$ replaced by $\mathfrak{B}_{\mathcal{D}}$; for $\mathcal{D}_\delta$ this adds the variables $\varsigma \in \mathbb{R}^S$ and the rows $-\varsigma \leq p - \tilde{p} \leq \varsigma, \mathbf{1}^\top \varsigma \leq 2\delta$. The objective F (eq. (F-def)) is unchanged.

Theorem 1 (Minimax reformulation under dependence; replaces Theorem 1). *Under Assumptions 1–3 and 4', let $\lambda^U \geq \bar{\lambda}$ (Lemma 2). Then*

$$\min_{x \in \mathcal{X}} V^*(x) = \min_{x \in \mathcal{X}} \max_{(\alpha, \zeta) \in \Omega_{\mathcal{D}}} F(x, \alpha, \zeta),$$

the inner maximum equals $V^*(x)$ for every x , and (i) $\Omega_{\mathcal{D}}$ does not depend on x and F is affine in x ; (ii) $\Omega_{\mathcal{D}}$ and F are linear in μ and bilinear in α and (b, t) , and fixing either α or (b, t) leaves a linear program. Consequently V^* is the restriction to $\mathcal{X}$ of a convex function on $\text{conv}(\mathcal{X})$, and every $(\alpha, \zeta) \in \Omega_{\mathcal{D}}$ yields an affine minorant $F(\cdot, \alpha, \zeta)$ of V^* on $\mathcal{X}$.

Proof. Fix x , $\alpha \in H$ and $b = (p, \nu, \tilde{p}) \in \mathfrak{B}_{\mathcal{D}}$. In $\Omega_{\mathcal{D}}$ the leader-block constraints (Om-L) involve b only through ν , and the follower-block constraints (Om-F)–(Om-lambda) only through $\tilde{p}$; $F = F^L + F^F$ splits accordingly. Two facts are established in the manuscript (proof of Theorem 1 and of Proposition 1(iii)):

- (a) with ν fixed, the leader block decomposes by scenario into the linear programming duals of (phi-fixed), with McCormick linearization of $x_i \pi$, scaled by ν_s ; its optimal value is $\sum_s \nu_s \phi(x, \alpha, \xi^s)$ (for $\nu_s = 0$ the scaled dual is homogeneous and its value is 0 because the primal is feasible);
- (b) with $\tilde{p}$ fixed, the follower block is the linear programming dual of (app-VF-LP) with the worst-case expectation replaced by the expectation under $\tilde{p}$, and its optimal value is $\mathfrak{F}_{\lambda^U}(x, \alpha; \tilde{p}) = -\lambda^U \text{gap}_x(\tilde{p}, \alpha)$ by (6).

Both blocks are feasible bounded linear programs, so their duals attain. Therefore

$$\begin{aligned} \sup_{(\alpha, \zeta) \in \Omega_{\mathcal{D}}} F(x, \alpha, \zeta) &= \sup_{\alpha \in H} \sup_{b \in \mathfrak{B}_{\mathcal{D}}} \left[\nu^\top \phi_x(\alpha) - \lambda^U \text{gap}_x(\tilde{p}, \alpha) \right] \\ &= \sup_{\alpha \in H} \sup_{(p, \tilde{p}) \in \mathcal{D}} \left[\mathcal{R}^p[\phi_x(\alpha)] - \lambda^U \text{gap}_x(\tilde{p}, \alpha) \right] = V_{\lambda^U}^*(x), \end{aligned}$$

using (3) for the middle equality, and $V_{\lambda^U}^*(x) = V^*(x)$ by Lemma 2(iii). The supremum is attained: $\alpha \mapsto \max_{b \in \mathfrak{B}_{\mathcal{D}}} \Phi_{\lambda^U}(\alpha, b)$ is continuous on the compact H , and for fixed α the remaining problem is a linear program over a polyhedron with finite value. Properties (i)–(ii) are those of the manuscript's Ω , since $\Omega_{\mathcal{D}}$ differs from Ω only by linear constraints on b , which involve neither x nor α . The consequences follow as in the manuscript. $\square$

Remark 3. This proof does not dualize the total-variation balls and does not use Sion's theorem, so it is shorter than the manuscript's proof also in the product case; the manuscript's derivation through (app-TV-dual) can be kept as the explicit construction of Ω , with the remark that the constraints on the multipliers $(p, \tilde{p})$ may be replaced by any polyhedral description of $\mathcal{D}$.

5 Follower types

Recall from the supplement (proof of Theorem 2) the index sets $I \subseteq \{1, \dots, m\}$ of rows of the lifted polytope $P_x = \{(h, \vartheta) : G(h, \vartheta) \leq g_x\}$, the faces $\Phi_I(x)$, and the x -independent cones $N_I = \{\tilde{p} \in \Delta^S : \exists \mu \geq 0, G_I^\top \mu = (0, \tilde{p})\}$. The Separation lemma of the supplement states $\alpha \in \Psi(x, \tilde{p})$ iff $\alpha \in \Phi_I(x)$ and $\tilde{p} \in N_I$ for some I , and its proof shows $\Phi_I(x) \subseteq \Psi(x, \tilde{p})$ for every $\tilde{p} \in N_I$.

Theorem 2 (Finite-type representation under dependence; replaces Theorem 2). *Under Assumptions 1–3 and 4', let $\mathcal{I}_{\mathcal{D}} := \{I : N_I \cap \text{proj}_{\tilde{p}} \mathcal{D} \neq \emptyset\}$ and, for $I \in \mathcal{I}_{\mathcal{D}}$,*

$$\hat{\mathcal{D}}_I := \text{proj}_p(\mathcal{D} \cap (\Delta^S \times N_I)),$$

a nonempty polytope. Both $\mathcal{I}_{\mathcal{D}}$ and $\hat{\mathcal{D}}_I$ are independent of x , and for every $x \in \mathcal{X}$:

- (i) $\Psi(x, \text{proj}_{\tilde{p}} \mathcal{D}) = \bigcup_{I \in \mathcal{I}_{\mathcal{D}}} \Phi_I(x)$; for $\mathcal{D}_\delta$ this equals $\bigcup_{\tilde{p} \in \mathcal{P}_{\tilde{\varepsilon}'}} \Psi(x, \tilde{p})$ with the finite type set $\mathcal{P}_{\tilde{\varepsilon}'}$ of the manuscript at radius $\tilde{\varepsilon}' = \min\{\tilde{\varepsilon}, \tilde{\varepsilon} + \delta\}$;

(ii) the leader's worst-case value is

$$V^*(x) = \max_{I \in \mathcal{I}_{\mathcal{D}}, \Phi_I(x) \neq \emptyset} \max_{\alpha \in \Phi_I(x)} \max_{p \in \mathcal{D}_I} \mathcal{R}^p[\phi(x, \alpha, \xi)], \quad (9)$$

with all maxima attained; for $\mathcal{D}_\delta$, $\hat{\mathcal{D}}_I = \hat{\mathcal{D}} \cap ((N_I \cap \tilde{\mathcal{D}}) \oplus B_\delta^{\text{TV}})$;

(iii) $V^*(x)$ is nondecreasing in δ , and type I is admitted ($I \in \mathcal{I}_{\mathcal{D}_\delta}$) exactly when $\gamma_I \leq \tilde{\varepsilon}'$, where $\gamma_I = \min\{d_{\text{TV}}(\tilde{p}, \hat{q}) : \tilde{p} \in N_I\}$ is the manuscript's threshold.

Proof. By Proposition 1 and the Separation lemma,

$$\begin{aligned} V^*(x) &= \max \left\{ \mathcal{R}^p[\phi_x(\alpha)] : (p, \tilde{p}) \in \mathcal{D}, \exists I : \alpha \in \Phi_I(x), \tilde{p} \in N_I \right\} \\ &= \max_I \max_{\alpha \in \Phi_I(x)} \max \left\{ \mathcal{R}^p[\phi_x(\alpha)] : (p, \tilde{p}) \in \mathcal{D}, \tilde{p} \in N_I \right\}, \end{aligned}$$

where for the converse inclusion we use that every $\tilde{p} \in N_I$ satisfies $\Phi_I(x) \subseteq \Psi(x, \tilde{p})$. The inner maximum is over $p \in \hat{\mathcal{D}}_I$, which proves (ii); $\hat{\mathcal{D}}_I$ is the projection of a polytope and does not involve x . The sets $\Phi_I(x)$ are faces' projections, hence compact, and $\mathcal{R}^p[\phi_x(\alpha)]$ is continuous, so the maxima are attained. (i) is the same computation without the objective, and for $\mathcal{D}_\delta$ it is Lemma 1(b) combined with the manuscript's Theorem 2(i) at radius $\tilde{\varepsilon}'$. (iii): $\mathcal{D}_\delta$ is nondecreasing in δ ; and $N_I \cap \text{proj}_{\tilde{p}} \mathcal{D}_\delta = N_I \cap \tilde{\mathcal{D}}_{\tilde{\varepsilon}'} \neq \emptyset$ iff $\gamma_I \leq \tilde{\varepsilon}'$. $\square$

Remark 4 (Types now carry their own leader set). In the product case $\hat{\mathcal{D}}_I = \hat{\mathcal{D}}$ for all admitted I , and one representative belief $\tilde{p}_I$ per type suffices. Under dependence each type carries the polytope $\hat{\mathcal{D}}_I$ of truths compatible with that type, so the representation is a robust game against finitely many follower types in which a type also restricts the data-side ambiguity. Between consecutive admission thresholds the sets $\hat{\mathcal{D}}_I$ grow continuously with δ , so $\delta \mapsto V^*(x)$ combines jumps (new types) with continuous growth.

6 Complexity

Theorem 3 (Σ_2^p -hardness under dependence; extends Theorem 3 and Corollary 1). *For every fixed $\tilde{\varepsilon} \in (0, 1]$ and $\delta \in (0, 1]$, the decision version of (2) with $\mathcal{D} = \mathcal{D}_\delta$ is Σ_2^p -hard already in the zero-sum case with $\hat{\varepsilon} = 0$, and evaluating $V^*(\bar{x})$ at a fixed $\bar{x}$ is NP-hard.*

Proof. With $\hat{\varepsilon} = 0$, $\hat{\mathcal{D}} = \{\hat{q}\}$ and, by Lemma 1(b), $\mathcal{D}_\delta = \{\hat{q}\} \times \tilde{\mathcal{D}}_{\min\{\tilde{\varepsilon}, \delta\}}$, a product set with follower radius $\min\{\tilde{\varepsilon}, \delta\} \in (0, 1]$. The manuscript's Theorem 3 and Corollary 1 hold for every fixed follower radius in $(0, 1]$. $\square$

Remark 5. For $\delta = 0$ (common ambiguous prior) and $\hat{\varepsilon} = 0$ there is no ambiguity at all; for $\delta = 0$ and $\hat{\varepsilon} > 0$ the reduction of Theorem 3 does not apply, because a corner belief would also be the leader's evaluation measure. The complexity of this case is open.

Proposition 2 (Saturation under dependence). *In the zero-sum case ($\phi = Q$) with a risk-neutral leader and $\hat{\varepsilon} \leq \tilde{\varepsilon}$, $V^*(x) = \max_{p \in \hat{\mathcal{D}}} \max_{h \in H} \mathbb{E}^p[Q(x, h, \xi)]$ for every $\delta \geq 0$.*

Proof. ($\leq$) For feasible $(p, \tilde{p}, \alpha)$, $\mathbb{E}^p[Q(x, \alpha, \xi)] \leq \max_h \mathbb{E}^p[Q(x, h, \xi)]$. ($\geq$) For $p \in \hat{\mathcal{D}} \subseteq \tilde{\mathcal{D}}$, the pair (p, p) lies in $\mathcal{D}_\delta$ and any $\alpha \in \Psi(x, p)$ attains $\max_h \mathbb{E}^p[Q]$. $\square$

7 Algorithm

All components of Section 4 of the manuscript are used with $\Omega_{\mathcal{D}}$ in place of Ω .

- **Master problem and cuts.** Unchanged: validity of $\ell_{(\alpha, \zeta)} = F(\cdot, \alpha, \zeta)$ needs only $(\alpha, \zeta) \in \Omega_{\mathcal{D}}$ and Theorem 1.
- **Local-optimality and restricted ISP cuts.** Unchanged; the restricted ISP (fixed α) is a linear program with S more variables and $2S + 1$ more rows for $\mathcal{D}_{\delta}$.
- **Lemma (price bounds).** Unchanged: its proof (Vertex-certificates lemma of the supplement) only modifies (ϖ, t) .
- **Spatial branch-and-bound.** The polytope $\Gamma(b, t) \leq \gamma$ now contains the rows of $\mathfrak{B}_{\mathcal{D}}$; for $\mathcal{D}_{\delta}$ the coupling rows $\varsigma_s - p_s + \tilde{p}_s \geq 0$, $\varsigma_s + p_s - \tilde{p}_s \geq 0$, $2\delta - \mathbf{1}^\top \varsigma \geq 0$ and the bounds of ς . (R1) multiplies them by the box factors of every α_i , introducing the lifted products $[\alpha_i \varsigma_s]$; (R2) is unchanged and may also be applied to $\varsigma \geq 0$. The relaxation remains valid and becomes exact as boxes shrink, so the proof of Proposition 2 (finite termination) is unchanged.

Proposition 3 (Belief cuts under dependence; replaces Proposition 1). *Under the assumptions of Theorem 1, with $v(x; \bar{b}, \bar{t}) := \max\{F(x, \alpha, \zeta) : (\alpha, \zeta) \in \Omega_{\mathcal{D}}, b = \bar{b}, t = \bar{t}\}$:*

- (i) (Validity) $v(x; \bar{b}, \bar{t}) \leq V^*(x)$ for every $\bar{b} \in \mathfrak{B}_{\mathcal{D}}$, prices $\bar{t}$ and $x \in \mathcal{X}$, and every optimal solution yields a valid cut.
- (ii) (Exactness) If $(\bar{\alpha}, \bar{b}, \bar{t}, \bar{\mu}) \in \Omega_{\mathcal{D}}$ then $v(\bar{x}; \bar{b}, \bar{t}) \geq F(\bar{x}, \bar{\alpha}, \bar{b}, \bar{t}, \bar{\mu})$; in particular $v(\bar{x}; b^*, t^*) = V^*(\bar{x})$ for the beliefs and prices of an optimal ISP solution.
- (iii) (Finiteness) Let $\mathfrak{B}_{\mathcal{D}, I} := \{b \in \mathfrak{B}_{\mathcal{D}} : \tilde{p} \in N_I\}$ (a polytope) and let $\mathcal{P}_{\mathcal{D}}^*$ be the set of pairs (b, t) with $b \in \text{vert}(\mathfrak{B}_{\mathcal{D}, I})$ for some $I \in \mathcal{I}_{\mathcal{D}}$ and $t_s = \theta^U \nu_s B^\top \sigma_s$ for vertices σ_s of Σ_F . Then $\mathcal{P}_{\mathcal{D}}^*$ is finite and independent of x , and $V^*(x) = \max_{(b, t) \in \mathcal{P}_{\mathcal{D}}^*} v(x; b, t)$ for every $x \in \mathcal{X}$.

Proof. (i)–(ii) are verbatim the manuscript’s proof, with Theorem 1 in place of Theorem 1. (iii) Finiteness and x -independence follow from Theorem 2 (N_I , $\mathcal{I}_{\mathcal{D}}$ do not depend on x). Fix x and let $(\alpha^*, p^*, \tilde{p}^*)$ attain (5); by the Separation lemma there is I^* with $\alpha^* \in \Phi_{I^*}(x)$ and $\tilde{p}^* \in N_{I^*}$, and $\nu^* \in L_{\beta}(p^*)$ attains $\mathcal{R}^{p^*}[\phi_x(\alpha^*)]$. Maximize the linear function $b \mapsto \nu^\top \phi_x(\alpha^*)$ over the polytope $\mathfrak{B}_{\mathcal{D}, I^*}$; a maximizing vertex $b^\circ = (p^\circ, \nu^\circ, \tilde{p}^\circ)$ satisfies

$$V^*(x) = \nu^{*\top} \phi_x(\alpha^*) \leq \nu^{\circ\top} \phi_x(\alpha^*) \leq \mathcal{R}^{p^\circ}[\phi_x(\alpha^*)] \leq V^*(x),$$

the last step because $(p^\circ, \tilde{p}^\circ) \in \mathcal{D}$ and $\alpha^* \in \Phi_{I^*}(x) \subseteq \Psi(x, \tilde{p}^\circ)$. Fix $b = b^\circ$ in $\Omega_{\mathcal{D}}$ with $\alpha = \alpha^*$: by facts (a)–(b) in the proof of Theorem 1, the leader block attains $\nu^{\circ\top} \phi_x(\alpha^*) = V^*(x)$ and the follower block attains $-\lambda^U \text{gap}_x(\tilde{p}^\circ, \alpha^*) = 0$. The Vertex-certificates lemma of the supplement replaces the certificates and prices by vertex-induced ones without decreasing $F(x, \cdot)$, which produces $(b^\circ, t^\circ) \in \mathcal{P}_{\mathcal{D}}^*$ with $v(x; b^\circ, t^\circ) \geq V^*(x)$; (i) gives equality. $\square$

8 Numerical verification

The implementation and all numerical checks are recorded in `docs/dependent_ambiguity/results.md`. In summary:

- **Exactness of $\Omega_{\mathcal{D}}$ and of the penalty λ^U (Theorem 1, Lemma 2).** At fixed x , the general $\Omega_{\mathcal{D}}$, the manuscript’s zero-sum Ω with the coupling rows, and a big-M-free KKT evaluation agree to $\leq 3 \times 10^{-7}$ (relative) on all tested (x, δ) pairs (zero-sum grid 3×3 , location instance).
- **Validity of all algorithmic components under a binding coupling** (location SGB128, $\delta = 0.1$, where $V_{0.1}([1]) = -893.75 < V_{\text{rect}}([1]) = -871.67$): every belief cut, restricted cut, Magnanti–Wong cut and Gurobi cut satisfies $\text{cut}(x) \leq V_{\delta}^*(x)$ at the checked points (up to

solver tolerance); the α -B&B bounds bracket the KKT value; belief cuts are exact at their generating point (3×10^{-11}).

- **Zero-sum code.** `true_dro_benders_optimize!` (mini-Benders replaced by the α -fixed joint LP under coupling) and `belief_menu_benders_optimize!` reach the same optimum as the validated general code on a grid instance with a binding coupling, for $\delta \in \{0.05, 0, \infty\}$.
- **Numerics.** Node LPs of the α -B&B can become ill-conditioned in very narrow α boxes (bound-factor RLT pairs become nearly dependent); the coupling only deepens the tree. A retry with `NumericFocus=3` resolves all observed cases.

9 Integration checklist for the manuscript

1. Assumption 4: state the dependent set (Assumption 4') and keep the product set as the case $\delta \geq \hat{\varepsilon} + \tilde{\varepsilon}$; delete “the dependent case is left to future work”.
2. Replace (V-response) and its paragraph by Proposition 1; the attainment argument of the supplement’s response-set section becomes the compactness argument in its proof.
3. Lemma 2: define $V_{\lambda_U}^*$ by (7); replace the proof of (iii) by Steps 1–4 of Lemma 2.
4. Theorem 1: replace $\mathfrak{B}$ by $\mathfrak{B}_{\mathcal{D}}$ in the definition of Ω (one line); the proof may be shortened as in Theorem 1.
5. Theorem 2: restate (ii) as (9); (i) holds at the effective radius $\tilde{\varepsilon}'$.
6. Theorem 3 and Corollary 1: add “for every fixed $\delta \in (0, 1]$ ” and the one-line reduction.
7. Proposition 1(iii): replace the independent choice of (p, ν) and $\tilde{p}$ by the joint vertex of $\mathfrak{B}_{\mathcal{D}, I}$.
8. Section 4.3: add the coupling rows to Γ .
9. Prose: the separation statements (abstract, Section 3 opening, after Lemma 2, after Theorem 2) become “the follower-side set determines the admissible responses, and each response restricts the leader-side set”.